C++ STL CHEAT SHEET (DSA + Interviews)

1. CONTAINERS

1.1 Vector
```
vector<int> v;
```
- push_back(x)
- pop_back()
- insert(v.begin()+i, x)
- erase(v.begin()+i)
- v[i]
- front(), back()
- size(), empty(), clear()

Note: Dynamic Array

1.2 Deque
```
deque<int> dq;
```
- push_back(x)
- push_front(x)
- pop_back()
- pop_front()
- front(), back()
- dq[i]
- size(), empty()

Note: Insert/Delete from both ends

1.3 Set
```
set<int> s;
```
- insert(x)
- erase(x)
- find(x)
- lower_bound(x)
- upper_bound(x)
- size(), empty(), clear()

Note: Sorted + Unique

1.4 Multiset
```
multiset<int> ms;
```
- insert(x)
- erase(x) // all x
- erase(ms.find(x)) // one x
- find(x)
- count(x)
- lower_bound(x)
- upper_bound(x)

Note: Sorted + Duplicates

1.5 Unordered Set
```
unordered_set<int> us;
```
- insert(x)
- erase(x)
- find(x)
- size(), empty()

Note: Unique, Not Sorted

1.6 Map
```
map<int,int> mp;
```
- mp[key] = value
- emplace(key,value)
- find(key)
- erase(key)
- size(), empty(), clear()

Note: Sorted by Key

1.7 Multimap
```
multimap<int,int> mm;
```
- insert({1,10})
- insert({1,20})
- find(1)
- count(1)
- erase(1)

Note: Duplicate Keys Allowed

1.8 Unordered Map
```
unordered_map<int,int> ump;
```
- ump[key] = value
- emplace(key,value)
- find(key)
- erase(key)
- size(), empty()

Note: Not Sorted

1.9 Frequency Count
```
unordered_map<char,int> freq;
for(char c : s)
    freq[c]++;
```
(Useful for counting)

2. ADAPTERS (CONTAINER ADAPTERS)

2.1 Stack
```
stack<int> st;
```
- push(x)
- pop()
- top()
- size(), empty()

2.2 Queue
```
queue<int> q;
```
- push(x)
- pop()
- front(), back()
- size(), empty()

2.3 Priority Queue

Max Heap (Default)
```
priority_queue<int> pq;
```
- push(x)
- pop()
- top()
- size(), empty()

Min Heap
```
priority_queue<int, vector<int>, greater<int>> pq;
```

Pair Priority Queue
```
priority_queue<pair<int,int>> pq;
```

3. PAIRS & TUPLES

3.1 Pair
```
pair<int,int> p;
```
- p.first
- p.second
- swap(p.first, p.second)

3.2 Tuple
```
tuple<int,char,double> t;
```
- get<0>(t)
- get<1>(t)
- get<2>(t)

4. VECTOR OF PAIRS
```
vector<pair<int,int>> vp;
```
- push_back({a,b})
- emplace_back(a,b)
- vp[i].first
- vp[i].second

Sort by First
```
sort(vp.begin(), vp.end());
```

Sort by Second
```
bool cmp(pair<int,int>& a,
         pair<int,int>& b){
  return a.second < b.second;
}
sort(vp.begin(),
     vp.end(), cmp);
```

Descending
```
sort(vp.rbegin(), vp.rend());
```

5. ITERATORS & LOOPS

5.1 Iterators
```
auto it = v.begin();
auto it = v.end();
```

5.2 Range Loop
```
for(auto x : v)
    cout << x;
```

6.3 By Reference
```
for(auto &x : v)
    cin >> x;
```

5.4 Iterator Loop
```
for(auto it=v.begin();
  it!=v.end(); it++){
  cout << *it;
}
```

6. ALGORITHMS

6.1 Sorting

Ascending
```
sort(v.begin(),
  v.end());
```

Descending
```
sort(v.begin(),
  v.end(),
  greater<int>());
```

Lambda Comparator
```
sort(vp.begin(), vp.end(),
  [](auto &a, auto &b){
    return a.second < b.second;
  });
```

6.2 Binary Search
```
binary_search(v.begin(),
  v.end(), x);
```
(Array must be sorted)

6.3 Lower / Upper Bound
```
auto lb = lower_bound(v.begin(), v.end(), x);
auto ub = upper_bound(v.begin(), v.end(), x);
```
- lower_bound → first element >= x
- upper_bound → first element > x

Index
```
int idx = lower_bound(v.begin(),
  v.end(), x) - v.begin();
```

6.4 Min / Max Element
```
auto mx = max_element(v.begin(), v.end());
auto mn = min_element(v.begin(), v.end());
```
Value:
```
*mx;    *mn;
```

6.5 Count
```
count(v.begin(), v.end(), x);
```

6.6 Find
```
find(v.begin(), v.end(), x);
```

6.7 Reverse
```
reverse(v.begin(), v.end());
```

6.8 Fill
```
fill(v.begin(), v.end(), x);
```

6.9 Accumulate
```
accumulate(v.begin(),
  v.end(), 0);
```

6.10 Next / Prev Permutation
```
next_permutation(v.begin(), v.end());
prev_permutation(v.begin(), v.end());
```

7. UTILITY FUNCTIONS

7.1 Min / Max
```
min(a, b);
max(a, b);
```

7.2 Absolute Value
```
abs(x);
```

7.3 String Stream
```
stringstream ss;
ss << "123";
ss >> x;
```

7.4 Swap
```
swap(a, b);
```

8. STRING FUNCTIONS

Size	s.size(); s.length();
Access	s[i]; s.front(); s.back();
Substring	s.substr(start, len);
Find	s.find("abc"); s.find(ch); // Not found: string::npos
Erase	s.erase(pos, len);
Insert	s.insert(pos, str);
Replace	s.replace(pos, len, newStr);
Reverse String	reverse(s.begin(), s.end());
Sort String	sort(s.begin(), s.end());
Case Conversion	toupper(ch); tolower(ch); (Entire String: for(char &c : s) c = toupper(c);)
Conversion	stoi(str); stol(str); stoll(str); stof(str); stod(str); to_string(x);
Input with Spaces	getline(cin, s);

9. TOP 15 FUNCTIONS IN DSA

1. sort()
2. reverse()
3. lower_bound()
4. upper_bound()
5. binary_search()
6. find()
7. count()
8. accumulate()
9. max_element()
10. min_element()
11. next_permutation()
12. fill()
13. swap()
14. stoi()
15. getline()